

Randomized and low-rank approximation

The Goreinov–Tyrtysnikov–Zamarashkin conjecture on square-submatrix inverse norms

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Improving the classical bound is challenging and would sharpen stability guarantees for skeleton/CUR approximation.

Problem statement

For integers $1 \leq r < n$, let $U \in \mathbb{R}^{n \times r}$ satisfy $U^T U = I_r$. Write $U_I = U(I, :)$ for $I \subseteq \{1, \dots, n\}$ with $|I| = r$, and let $\|\cdot\|_2$ denote the spectral norm.

Conjecture (Goreinov–Tyrtysnikov–Zamarashkin [1, Eq. (2.6)]). Every such U admits a nonsingular square submatrix U_I satisfying

$$\|U_I^{-1}\|_2 \leq \sqrt{n}.$$

In the notation of [1], this is $t(r, n) \leq \sqrt{n}$, where

$$t(r, n) = \max_{\substack{U \in \mathbb{R}^{n \times r} \\ U^T U = I_r}} \min_{\substack{I \subseteq \{1, \dots, n\} \\ |I| = r}} \|U_I^{-1}\|_2.$$

Known bounds and cases

The bound

$$t(r, n) \leq \sqrt{r(n-r) + 1}$$

is well known and follows from various arguments: maximal volume [1], Bischof–Stewart QR (BSQR) [2, 3], or volume sampling [4, 5] just to name a few. Also, $t(r, n) \geq \sqrt{n}$ whenever $r+1$ divides n [1, Lem. 2.2], so the conjectured constant is sharp.

The real case $r = 1$ is elementary. Nesterenko [6] proved the real case $n = 4, r = 2$; Sengupta–Pautov [7] subsequently proved $r = 2$ for all n . Orthogonal completion gives $t(r, n) = t(n-r, n)$, leaving $3 \leq r \leq n-3$ as the general open range. These inverse norms control pseudoskeleton approximation errors [1, Thms. 3.1–3.2].

Proposed complex extension

Define $t_{\mathbb{C}}(r, n)$ analogously using $U \in \mathbb{C}^{n \times r}$ with $U^* U = I_r$, where $*$ denotes conjugate transpose. The classical bound $\sqrt{r(n-r) + 1}$ also holds over $\mathbb{C}$ [3].

Conjecture. There is an absolute constant α , independent of r and n , such that

$$t_{\mathbb{C}}(r, n) \leq \alpha \sqrt{n}, \quad 1 \leq r < n.$$

The dependence on r, n and the best possible α remain to be determined. The constant $\alpha = 1$ fails: $t_{\mathbb{C}}(2, 4) = \sqrt{3 + \sqrt{3}} > 2$ [8, 9]. For two columns, Nesterenko [9, Prop. 1] proves

$$t_{\mathbb{C}}(2, n) \leq c_2 \sqrt{n}, \quad c_2 = \left(2 - \frac{2}{\sqrt{3}}\right)^{-1/2} \approx 1.08766,$$

with equality whenever $4 \mid n$, so any universal α must satisfy $\alpha \geq c_2$.

References

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7. R. Sengupta and M. Pautov, *On the submatrices with the best-bounded inverses*, arXiv: 2604.05944v5 (2026). [Preprint](#).
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Checked [1]–[9] and searched for later resolutions; no general real-case resolution or resolution of the proposed complex extension was found. The real two-column result is a preprint and supports the partially resolved status.